

Assignment

1: Test Case Design & Review

a. Application Selected: Login Page

Objectives of Test Case Design

- To verify that the login functionality works correctly.
- To ensure valid users can log in successfully.
- To ensure invalid users are restricted from logging in.
- To validate input fields (username and password).
- To check error messages and security validations.

Scope of Test Case Design

- Testing username and password fields.
- Testing login button functionality.
- Testing validation messages.
- Testing positive and negative scenarios.
- Excludes backend database testing and performance testing.

b. Test Case Document

Features to be Tested

- Username input field validation
- Password input field validation
- Login button functionality
- Error message display
- Successful login redirection

Entry Criteria

- Application is developed and deployed in test environment.
- Test cases are prepared and reviewed.
- Test data is available.

Exit Criteria

- All test cases are executed.
- All critical defects are fixed and retested.
- Test summary report is prepared.

Preconditions

- User must be registered in the system.

- Application URL is accessible.
- Valid test credentials are available.

c. Test Cases (Positive & Negative Scenarios)

Test Case ID	Description	Steps	Test Data	Expected Result
TC_01	Login with valid credentials	Enter valid username & password Click Login	Username: user1 Password: Pass@123	User should login successfully
TC_02	Login with invalid password	Enter valid username & wrong password Click Login	user1 / wrong123	Error message displayed
TC_03	Login with invalid username	Enter invalid username & valid password Click Login	wronguser / Pass@123	Error message displayed
TC_04	Login with empty fields	Leave username & password blank Click Login	Blank	Validation message displayed

TC_05	Login with only username	Enter username only → Click Login	user1 / Blank	Password required message
TC_06	Login with special characters	Enter special characters in username/pass word	@@@ / ###	Proper validation message displayed

d. Two Key Benefits of Test Case Review

1. Error Identification Early – Helps identify missing test scenarios and incorrect test steps before execution.
2. Improves Quality & Coverage – Ensures all requirements are covered and reduces defect leakage.

2: Test Execution, Defect Life Cycle & Test Closure

a. Test Environment (Test Bed)

A Test Environment is a setup that includes hardware, software, network configuration, and test data required to execute test cases.

Importance:

- Ensures testing is done in a stable environment.
- Helps identify environment-related issues before production release.
- Ensures accurate and reliable test results.

b. Defect Life Cycle

Definitions:

- Mistake – Human action that produces incorrect result.
- Bug – Flaw in software code.
- Defect – Deviation from expected result found during testing.
- Error – Difference between actual and expected result.

Defect Life Cycle Stages:

1. New
2. Assigned
3. Open
4. Fixed
5. Retest
6. Verified
7. Closed

(If not fixed → Reopened)

c. Executed Test Cases

Test Case ID	Description	Expected Result	Actual Result	Status	Comments
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TC_01	Valid login	User logged in	User logged in	Pass	Working fine
TC_02	Invalid password	Error message	Error message displayed	Pass	Correct validation
TC_03	Invalid username	Error message	System allows login	Fail	Critical defect
TC_04	Empty fields	Validation message	Validation message displayed	Pass	Proper validation
TC_05	Only username entered	Password required message	No message displayed	Fail	Minor defect

d. Difference Between Test Summary Report & Test Signoff Document

- Test Summary Report – Document prepared at end of testing that summarizes test results, defects, and overall testing status.
- Test Signoff Document – Formal approval document stating testing is completed and application is ready for release.